

CAMPENI, Romania

WMO: 151620

Lat: 46.364N

Lon: 23.040E

Elev: 1943

StdP: 13.69

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.7	(c) 10.2	(d) 2.2	(e) 6.6	(f) 7.9	(g) 6.4	(h) 8.2	(i) 12.2	(j) 15.3	(k) 26.5	(l) 9.3	(m) 31.4	(n) 1.2	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.9	(c) 86.2	(d) 68.4	(e) 83.1	(f) 67.2	(g) 79.9	(h) 65.8	(i) 70.3	(j) 82.6	(k) 68.6	(l) 80.0	(m) 67.2	(n) 77.8	(o) 3.3	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	104.6	74.9	64.8	98.6	73.5	63.4	93.7	72.0	35.5	83.1	34.0	79.8	32.8	77.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.7	(b) 7.6	(c) 6.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.7	(b) 7.6	(c) 6.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	28.0	33.0	39.6	47.8	56.6	63.6	65.4	66.3	57.4	49.1	38.8	31.0		
	DBStd	14.89	8.17	6.94	7.15	7.96	5.39	5.90	4.80	5.20	6.75	6.83	7.44	7.13		
	HDD50	2671	683	476	328	133	12	0	0	0	13	100	338	588		
	HDD65	6376	1148	896	789	516	263	95	54	48	237	492	785	1053		
	CDD50	1990	0	0	4	67	215	408	478	506	236	73	3	0		
	CDD65	222	0	0	0	0	2	53	67	90	10	0	0	0		
	CDH74	2553	0	0	0	27	95	545	749	943	178	16	0	0		
	CDH80	685	0	0	0	0	9	129	204	312	31	0	0	0		
(14) Wind	WSAvg	1.9	1.4	2.0	2.1	2.5	2.2	2.1	1.9	1.8	1.6	1.5	1.3	1.8		
(15) Precipitation	PrecAvg	31.9	1.7	1.5	1.9	2.7	3.4	4.0	4.2	3.0	2.8	2.4	2.2	2.4		
	PrecMax	41.1	3.5	4.1	5.2	5.1	6.1	7.1	7.0	7.5	6.2	5.6	3.9	5.4		
	PrecMin	22.1	0.5	0.3	0.5	0.4	1.9	1.3	1.2	0.7	0.4	0.1	0.0	0.2		
	PrecStd	5.4	0.8	0.9	1.3	1.1	1.2	1.5	1.6	1.3	1.5	1.5	0.9	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	56.4	69.1	77.6	80.7	87.3	89.3	91.3	84.4	75.8	61.4	48.5		
		MCWB	41.9	47.7	52.8	58.8	62.1	70.8	68.8	69.2	64.3	57.9	52.7	44.0		
	2%	DB	42.9	51.1	62.6	73.4	76.8	83.7	85.0	86.7	79.1	71.2	55.6	44.5		
		MCWB	39.6	43.5	50.1	56.5	62.0	68.4	68.5	67.6	63.7	57.0	49.4	42.2		
	5%	DB	39.9	46.3	56.5	69.0	73.0	80.5	81.5	83.2	75.1	66.3	52.1	41.8		
		MCWB	37.6	40.9	47.0	54.0	60.1	67.1	67.4	66.8	61.7	55.5	48.0	39.9		
	10%	DB	37.3	42.3	51.5	63.7	69.1	76.5	78.1	79.0	70.3	60.8	48.8	39.0		
		MCWB	35.8	38.6	43.8	51.7	58.5	65.0	66.1	66.1	60.4	53.4	46.2	37.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	48.3	54.5	60.3	65.6	71.9	72.8	71.6	67.0	60.7	54.0	45.5		
		MCDB	46.3	56.6	67.9	74.8	78.0	85.5	83.0	86.3	80.9	70.8	58.8	46.8		
	2%	WB	40.2	44.7	50.6	57.7	62.7	69.7	70.1	69.8	64.8	58.3	51.1	42.2		
		MCDB	42.0	49.8	61.1	71.0	73.8	81.5	81.3	83.3	76.6	68.9	54.8	44.1		
	5%	WB	38.0	41.6	47.8	55.2	60.9	67.8	68.4	68.1	62.8	56.1	48.9	39.8		
		MCDB	39.7	44.9	55.8	67.6	70.4	78.7	79.3	79.3	72.3	64.8	51.6	41.4		
	10%	WB	36.0	39.1	44.9	52.7	59.1	66.0	67.0	66.7	61.0	53.7	46.6	37.8		
		MCDB	37.2	41.6	50.6	62.6	67.8	75.2	76.7	77.6	69.2	59.8	48.6	38.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	17.2	20.1	23.4	22.0	22.7	23.9	24.5	21.3	22.7	15.8	11.9		
		MCDBR	15.7	24.1	31.0	33.7	29.1	28.3	29.5	30.9	29.2	31.9	21.6	14.0		
		MCWBR	11.9	15.8	18.1	17.3	14.9	13.7	14.0	13.0	13.5	17.9	14.4	10.7		
	5% WB	MCDBR	13.9	20.7	28.7	31.0	26.2	26.7	26.5	27.7	25.9	29.2	18.4	13.5		
		MCWBR	10.6	14.2	17.2	16.0	13.9	13.3	13.5	12.6	12.6	16.5	13.0	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.342	0.388	0.442	0.427	0.420	0.439	0.446	0.406	0.369	0.340	0.314		
	taud		2.316	2.249	2.189	2.074	2.180	2.238	2.192	2.168	2.249	2.338	2.363	2.352		
	Ebn at Noon		244	259	262	258	267	269	262	254	254	246	232	230		
	Edh at Noon		24	32	39	49	45	43	44	44	37	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		387	640	1003	1375	1654	1833	1796	1656	1178	811	467	307		
	RadStd		53	82	145	143	157	158	158	160	132	96	63	40		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+1.28	N/A	N/A	N/A	+1.80	N/A	N/A	-225	-340	+218	+108

Nomenclature: See separate page